
Estimating Active Dimension of Training Dynamics from One Scalar Log, with an Application to Grokking

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Abstract

Over a stretch of training in which the update rule is fixed and the parameters keep returning near states already visited, the states they revisit form a set of dimension far below the number of parameters: the active dimension of that regime. Measuring it needs the stored trajectory, whereas a run already writes scalar logs: how much of that dimension survives in one such scalar? We reconstruct a state space by delay coordinates and estimate its intrinsic dimension by maximum likelihood on nearest-neighbour distances. The procedure returns a plausible number on any record, including one in which nothing recurs, so it estimates the active dimension only under conditions we establish separately. On the one of six constructed systems that meets every requirement we fixed in advance it is accurate to about one component, up to a ceiling near eight; The remaining constructions expose failure modes under transience, stochastic forcing and poor delay choices. All validation systems are externally forced; the training runs are not. The diagnostics therefore reject the scalar estimate as a dimension in both grokking settings. We instead analyse saved trajectory sketches. The detrended covariance participation ratio falls near generalisation in four regularised runs and later recovers. A memorising run can fall as far. The timing of the fall and its reversal separates the generalising runs in this sample. A matched short-window analysis of the parameter norm shows the same event, but its level is not dimensional and the effect is weaker than the corresponding change in roughness.

1 INTRODUCTION

A network is trained in a parameter space of dimension P , but over a stretch of training the parameters usually move in far fewer dimensions. Is an apparent simplification of training a real reduction in the number of degrees of freedom in use, or an artefact of a shrinking scale? Can a phase transition such as delayed generalisation be detected without instrumenting the run? Both are argued from a count of degrees of freedom. Those in use disagree: the *available* dimension of the subspace the parameters are confined to, which Li et al. (2018) take as the smallest that still solves the task, the *functional* dimension of the map from that subspace to the outputs, and the *effective rank* of the trajectory covariance.

Our two questions concern a fourth quantity: the number of independent components of the set the parameters occupy while the update rule is fixed. We call it the *active dimension* of that regime (equation (1)). If the motion is forced by r independent quasiperiodic phases, the path lies on an r -dimensional torus and the active dimension is r . Section 5.1 moves it while the available and functional dimensions stay fixed.

Measuring the active dimension means storing the parameter trajectory, which an ordinary run does not do. A run already writes scalar logs: a loss, a gradient norm, a parameter norm. *How much of the active dimension can be recovered from one such scalar?* Nonlinear time series analysis supplies a standard answer: reconstruct a state space by delay coordinates (Takens, 1981; Sauer et al., 1991), then estimate its intrinsic dimension by maximum likelihood on nearest-neighbour distances (Levina and Bickel, 2004; MacKay and Ghahramani, 2005).

The procedure returns a number on any record. Power-law noise yields a finite and reproducible value fixed by its spectral exponent (Osborne and Provenzale, 1989; Theiler, 1991); a decaying transient yields a large value. Establishing which case a training log falls in is the substance of the problem.

Contributions. We make three contributions.

1. **An estimator of active dimension from scalar logs.** We reconstruct delay coordinates from one scalar training log and apply a pooled nearest-neighbour estimator, together with diagnostics for recurrence and identifiability. The protocol is computationally light and avoids storing the full parameter trajectory. Under the delay-embedding conditions of theorems 1 to 3, a generic scalar observation preserves the relevant state geometry in delay coordinates, including fibre by fibre for forced systems. The contribution is this calibrated scalar-log protocol: it distinguishes changes in active components from changes in scale, anisotropy, roughness, or failed recurrence, without claiming novelty for delay embeddings or intrinsic-dimension estimators individually. A single mini-batch trajectory can cross forcing fibres, so we identify such cases as outside the dimensional interpretation.
2. **Controlled validation with independent ground truth.** We test the procedure on six constructed systems spanning analytic oscillators, optimisation dynamics, and a learned image-data system. Under a frozen held-out protocol, the estimate recovers the ground-truth rank to about one component in the admissible deterministic recurrent regime and orders withheld ranks correctly. The remaining systems expose failure under transience, stochastic forcing, unequal excitation, or insufficient delay coordinates, and show why covariance effective rank and log roughness are not substitutes for active dimension.
3. **Application to grokking with an explicit validity check.** In two grokking settings, the scalar logs fail the recurrence and identifiability tests, so their apparent dimension estimates are not interpreted as active dimension. Stored trajectory sketches nevertheless detect local simplification in grokking-type systems, including $WD = 0$ controls: the detrended covariance participation ratio drops and then recovers. The drop aligns with generalisation in regularised runs, whereas a $WD = 0$ control can show it elsewhere in training; the signal therefore detects simplification but does not by itself identify generalisation or prove a change in active dimension.

2 RELATED WORK

Reconstruction and intrinsic dimension. Delay reconstruction follows from Takens’ theorem and its extensions (Appendix A), is used on empirical series (Sugihara and May, 1990; Sugihara et al., 2012), and has standard rules for choosing its lag and dimension (Fraser and Swinney, 1986; Kennel et al., 1992; Cao,

1997; Broomhead and King, 1986). We use pooled maximum likelihood (Levina and Bickel, 2004; MacKay and Ghahramani, 2005); the alternatives (Grassberger and Procaccia, 1983; Facco et al., 2017; Camastra and Staiano, 2016) are bounded alike by a finite record (Eckmann and Ruelle, 1992). Oversampling (Theiler, 1986) and power-law noise (Osborne and Provenzale, 1989) produce spurious dimensions; surrogate tests (Theiler et al., 1992; Schreiber and Schmitz, 1996) detect non-linear structure but not how much is present and miss a decaying transient entirely.

Dimension in optimisation. Random-subspace training (Li et al., 2018; Aghajanyan et al., 2021) reports the smallest k that still solves a task, a property of the task and not of a window; we adopt the same parameterisation but measure a different quantity within it. Closest to the active dimension is the finding that descent occurs largely in a small subspace spanned by the top Hessian eigenvectors (Gur-Ari et al., 2018; Sagun et al., 2017). Ansuini et al. (2019) and Pope et al. (2021) estimate the intrinsic dimension of a data set or of a layer’s representations, properties of a point cloud rather than of a path. The participation ratio we adopt (Gao et al., 2017) is not the spectral-entropy quantity of the same name (Roy and Vetterli, 2007); it has been measured on a grokking trajectory once, in passing (Notsawo Jr. et al., 2023). All of these quantities need weights, gradients or activations, whereas the nearest precedent for a scalar computed during training and acted on is the gradient noise scale (McCandlish et al., 2018).

Grokking. Delayed generalisation (Power et al., 2022) is studied in two settings and we use both. The transformer setting is regularised and, in our version, stochastic (Nanda et al., 2023; Liu et al., 2023); the perceptron setting is the unregularised full-batch gradient descent of Gromov (2023), on the modular polynomials and representability criterion of Doshi et al. (2024), which supply label-matched pairs (Appendix J). Explanations are framed in weight or activation space (Nanda et al., 2023; Varma et al., 2023; Liu et al., 2022; Barak et al., 2022) or in a lazy-to-rich transition (Kumar et al., 2024). Closest to us, Notsawo Jr. et al. (2023) predict grokking from early loss oscillations (Thilak et al., 2022) and Merrill et al. (2023) track competing subnetworks. We differ from all of these in estimating a dynamical quantity whose ground truth is measured independently.

3 PROBLEM STATEMENT

3.1 The estimation problem

Training can appear to simplify: after a long period of memorisation, a network may begin to generalise while its parameter trajectory occupies fewer effective directions. The problem is to decide whether this apparent simplification reflects a genuine reduction in the number of independent components explored by the training dynamics, rather than a change in scale, unequal excitation of existing directions, transient motion, or stochastic forcing. The usual available dimension, functional dimension, and covariance effective rank answer related but different questions; none is, by itself, the local dimension of the recurrent regime visited by the optimiser.

Problem statement. Let R be a stretch of training with a fixed update rule and optimiser state $\mathbf{s}_t$. We observe only a scalar log $x_t = \phi(\mathbf{s}_t)$; the full state trajectory may be unavailable or too expensive to store. Given a finite window of x_t , we seek an estimator that (i) estimates the number of independent components of the regime’s occupation measure, and (ii) identifies when the available log and window do not support that interpretation. Formally, the target is the active dimension $d_{\text{act}}(R)$ defined in equation (1); the estimator should approximate this target in a recurrent regime and report its output as a dimension only when the validity conditions hold.

Target and estimator. Equation (1) defines the quantity we want to measure: the local scaling exponent of the regime’s occupation measure. We estimate this exponent from a finite record by reconstructing the state with delay coordinates and applying the pooled nearest-neighbour maximum-likelihood estimator of Levina and Bickel, written explicitly in equation (3). Thus equation (1) is the target definition, while equation (3) is the finite-sample estimator. The delay-embedding results of Takens, Sauer, and Stark explain why this can be done from a scalar observation: under their genericity and recurrence conditions, the delay map preserves the relevant state geometry (for a forced system, fibre by fibre), so neighbour distances in the reconstructed log can carry information about the target dimension. These theorems do not certify every training log: in particular, a stochastic mini-batch trajectory can cross different forcing fibres. The estimator therefore operates together with explicit identifiability and regime diagnostics, and returns a statistic without a dimensional interpretation when those conditions fail.

3.2 The active dimension

Let $\mathbf{s}_t = (\boldsymbol{\theta}_t, \mathbf{u}_t)$ be the optimiser state at step t : the parameter vector $\boldsymbol{\theta}_t \in \mathbb{R}^P$ and whatever the optimiser carries beside it. Where the parameters are confined by construction or by the dynamics to an affine subspace $\boldsymbol{\theta} = \boldsymbol{\theta}_0 + \mathbf{V}^\top \mathbf{c}$ with $\mathbf{V} \in \mathbb{R}^{k \times P}$ orthonormal, we work in the free coordinates $\mathbf{c}_t \in \mathbb{R}^k$; that change of coordinates is an isometry and preserves intrinsic dimension.

We call k the *available dimension*. The *functional dimension* is the rank of $\partial \mathbf{f}(\mathbf{c}) / \partial \mathbf{c}$, with $\mathbf{f}$ the outputs on a fixed probe set; it bounds how many of the k directions can change the computed function. The *active dimension*, the subject of this paper, is a property of a *regime* R : a stretch of training over which the update rule is fixed and the state carries an ergodic invariant measure. Carried to the free coordinates, that measure gives the *occupation measure* μ_R , the long-run fraction of steps at which $\mathbf{c}_t$ falls in a given set. The regime occupies the support $\mathcal{A}_R$ of μ_R , the points whose every neighbourhood is visited with positive frequency. The active dimension is the exponent with which μ_R fills a small ball about such a point,

$$d_{\text{act}}(R) = \lim_{\varepsilon \downarrow 0} \frac{\log \mu_R(B(\mathbf{c}, \varepsilon))}{\log \varepsilon}, \quad (1)$$

with $B(\mathbf{c}, \varepsilon)$ the ball of radius ε about $\mathbf{c}$. A measure whose exponent is the same at μ_R -almost every $\mathbf{c}$ is *exactly dimensional*; that common value is $d_{\text{act}}(R)$. Where $\mathcal{A}_R$ is a smooth d -dimensional manifold carrying a positive density the exponent is d ; a point of it is then fixed by d local coordinates, the *active components*.

No record reaches the limit. Between two radii the same ratio gives the finite-scale exponent $d_{\text{act}}(R; \varepsilon_1, \varepsilon_2)$ of equation (6), which agrees with $d_{\text{act}}(R)$ where the same power continues below ε_1 and otherwise counts only the phases forced above it (section 6.4). A window $\mathcal{W}$ of L steps samples μ_R rather than being the set: its own exponent is zero, and it returns that finite-scale exponent at the radii its neighbour distances span.

The systems of section 5 set $\mathbf{c}_t = F(\varphi_1(t), \dots, \varphi_r(t))$ with r rationally independent quasiperiodic phases and F a smooth embedding of the r -torus, giving $\mathcal{A}_R = F(\mathbb{T}^r)$ and $d_{\text{act}}(R) = r$. Rational independence makes the orbit dense on the torus; the embedding keeps F from merging or discarding a phase.

We also measure the participation ratio (Gao et al., 2017), which for a non-negative spectrum $\lambda_1, \dots, \lambda_n$ is

$$\text{PR}(\lambda) = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}. \quad (2)$$

It equals m exactly when m of the λ_i are equal and the rest vanish. Applied over a window to the covariance of $\mathbf{c}_t$ it gives the *effective rank* PR^{pos} and to the covariance

Table 1: A guide to the quantities used to describe training. The final column gives the limits of their interpretation.

Quantity	Meaning	Required data	Interpretation limit
Available dimension k	Number of parameter directions allowed by the subspace.	The subspace parameterisation.	Does not specify how many directions the trajectory explores.
Functional dimension	Rank of the output Jacobian at a given parameter state.	The Jacobian on a fixed probe set.	A lower rank need not imply a lower dimension of the parameter trajectory.
Active dimension $d_{\text{act}}(R)$	Local dimension of the regime's occupation measure (equation (1)).	A sampled recurrent trajectory; a suitable scalar log under the conditions of section 3.4.	A smaller scalar estimate alone does not establish a fall in this dimension.
Effective rank $\text{PR}^{\text{pos}}, \text{PR}^{\text{det}}$	Effective number of directions carrying the trajectory variance (equation (2)).	A trajectory window; detrended within the window for PR^{det} .	Can fall through unequal variances while active dimension stays fixed.

of the residual $\tilde{c}_t$ about the window's least-squares line the detrended form PR^{det} . A drifting trajectory needs the detrended form, since a straight drift dominates the spectrum and holds PR^{pos} near unity however the trajectory moves about that line. Written without a superscript, *the effective rank* means PR^{pos} .

The effective rank is not the active dimension: it falls below r wherever the r directions do not carry comparable variance, while $d_{\text{act}}(R)$ stays at r (section 6.4). Table 1 summarises the quantities and their interpretation; the dynamical regimes are listed in section 3.4.

3.3 Observation model and estimator

We observe a scalar series $x_t = \phi(\mathbf{s}_t)$, in which $\mathbf{s}_t$ is the full optimiser state and ϕ is fixed before the run. From it we reconstruct $\mathbf{y}_t = (x_t, \dots, x_{t-(E-1)\tau}) \in \mathbb{R}^E$. For a generic pair of dynamics and observation on a compact invariant set of box-counting dimension d , this map is an embedding once $E > 2d$ (theorems 1 and 2). Genericity is not verifiable for a ϕ fixed in advance, so every result below is reported over a family of observers. We

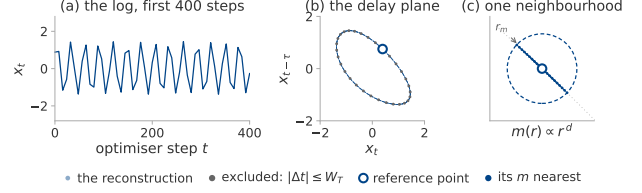


Figure 1: One scalar log carries the geometry the estimator counts. The one-phase case of section 5.1: (a) 400 of its 10 000 logged steps; (b) the reconstruction of the whole record in its first two delay coordinates, a closed loop; (c) the neighbours of one point and the radius r_m of equation (3). The record passes round the loop many times over, so the excluded samples lie all round it and every neighbour in (c) is a return from another pass.

estimate the intrinsic dimension of $\{\mathbf{y}_t\}$ by maximum likelihood on nearest-neighbour distances (theorem 4). Let $r_1^{(i)} \leq \dots \leq r_m^{(i)}$ be the distances from point i to its m nearest neighbours, counting a neighbour only if it is more than W_T steps away from i in time (figure 1). This is the *Theiler exclusion* (Theiler, 1986; Kantz and Schreiber, 2004): without it the nearest points in $\mathbb{R}^E$ are the samples just before and after i , so the statistic measures the sampling rate instead of the geometry. Pooling the likelihood over the N points before inverting, as MacKay and Ghahramani (2005) recommend, gives

$$\hat{d}_{\text{MG}}^{-1} = \frac{1}{N(m-1)} \sum_{i=1}^N \sum_{j=1}^{m-1} \log \frac{r_m^{(i)}}{r_j^{(i)}}, \quad (3)$$

Appendix B.1 gives the algorithm as implemented; table 10 scores the per-point average of Levina and Bickel (2004) beside it. The *roughness* $\text{std}(\Delta x)/\text{std}(x)$, the *linear participation ratio* PR_{delay} of equation (2) on $\text{Cov}\{\mathbf{y}_t\}$ and the *spectral participation ratio* on the power spectrum of x accompany every estimate as nulls: each can grow with r for reasons that involve no geometry, so an estimator no more accurate than the best of them has measured nothing.

3.4 Dynamical regimes

Before an estimate computed from a log gives an active dimension, the window must lie inside a single regime. The embedding theorems (theorems 1 and 2) require a deterministic map on a compact invariant set, which licenses the delay reconstruction as a change of coordinates. The neighbour statistic of equation (3) requires more: the record must cover that set densely enough that the Theiler exclusion still leaves each point a genuine return. Recurrence is needed because our entire sample is one trajectory. Last, the neighbour

radii must span a range over which the exponent is constant. Where a condition fails, $\hat{d}_{\text{MG}}$ is a statistic of the window and not a dimension.

- **Deterministic and recurrent.** The path returns close to states already passed through and lies on an r -dimensional torus, so $\mathcal{A}_R$ has r independent components and $d_{\text{act}}(R) = r$; the estimator recovers it.
- **Deterministic and transient.** A converging run decays along a curve it never revisits. The curve is one-dimensional, but a run that does not return samples no invariant measure, so it has no $\mathcal{A}_R$ and equation (1) has nothing to measure: the dimension of a path and the dimension of a regime are different quantities, and here the estimator returns a statistic of the window (section 6.1). Its effective rank does stay near unity (table 13), because the curve is nearly straight.
- **Stochastically forced.** A stationary mini-batch optimiser may have an occupation measure of its own, whose dimension counts the directions the gradient noise excites. That is not the quantity we are after. The embedding theorems for forced systems (theorem 3) do cover forcing of this kind, but one realisation of it at a time, whereas a single run crosses realisations.

3.5 Diagnostics

The series alone is enough to tell a transient from stochastic forcing. The **identifiability ratio** $\rho_{\text{ident}} = \hat{d}_{\text{MG}}(2E_{\text{max}})/\hat{d}_{\text{MG}}(E_{\text{max}})$ compares the estimate at a doubled embedding dimension with the estimate at E_{max} , the one a configuration fixes. The ratio stays near unity where a dimension can be measured and grows wherever the estimate follows the embedding dimension rather than the data: under mini-batch noise, and furthest of all on a transient (figure 2b). It does not say which of the two has happened. The **trend-crossing count** does, being the number of sign changes of the residual of the window about its own least-squares line, near zero on a monotone decay. Neither tests recurrence itself (Appendix I); nothing we compute tests the scaling range.

4 VALIDATION PROTOCOL

The requirements below were fixed before any result was computed; table 6 records which of them each system fails.

Constructed truth: the ground truth follows from the construction rather than from a fit. Every system is built with r independent quasiperiodic phases, so that $d_{\text{act}}(R) = r$; the effective rank of the recorded

trajectory must then confirm that all r are comparably excited.

Frozen configuration: the estimator is chosen once, by a search over its own settings alone. Searching over the system at the same time would select the data together with the method.

Withheld ranks: ranks are held out as well as seeds, because the geometry of the excitation depends on r and every seed at a given rank shares that geometry.

Zero learning rate: every system is re-run with the learning rate set to zero, so any observer of the optimiser must be constant. A log that still varies and still returns its earlier estimate was never a function of the optimiser.

5 EVALUATION ON SYSTEMS WITH A KNOWN ACTIVE DIMENSION

We evaluate the estimator on six systems of increasing realism, from an oscillating diagonal matrix to a linear head on a network trained on image data (table 6). The first has no optimiser and no feedback: in $\mathbf{W}(t) = \text{diag}(b_i + \delta_i(t))$ the offsets b_i are fixed and r of the δ_i oscillate at rationally independent frequencies; the estimate tracks r across the whole withheld range to within a third of a component.

5.1 A constrained head on a network trained on image data

A tanh network is trained on the scikit-learn handwritten digits and frozen; a linear head confined to a k -dimensional affine subspace is then trained on top of it under softmax cross-entropy, so the curvature is data-dependent and all the dynamics occur in $\mathbf{c}$. The excitation is made r -dimensional by modulating the loss weights of r of twelve fixed data groups (Appendix E). We force the head quasiperiodically, with mini-batch noise, and by plain gradient descent, covering the recurrent, stochastic and transient regimes of section 3.4.

Twelve scalar observers are recorded (Appendix B.2). The estimator was frozen on four of them, on seeds and ranks reserved for that purpose, so every result below is out of sample in rank, in seed, and for most observers in the observer too. In the deterministic recurrent regime the estimate recovers the active dimension to within about one component. Every observer orders the withheld ranks correctly, and none does so through the roughness of its log (table 8).

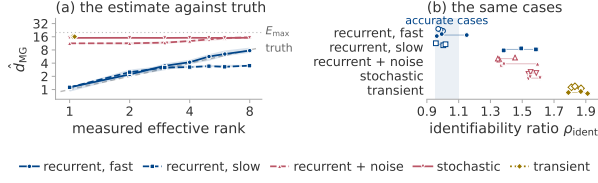


Figure 2: Recovery of the active dimension in each regime. The linear head of section 5.1 under five excitations, coloured by the regime of section 3.4: two quasiperiodic forcings at different rates, the fast one with gradient noise added, mini-batch noise alone, and plain gradient descent. (a) Estimate against truth: marker, median over seeds and observers; band, their interquartile range; dashed, exact recovery; dotted, $E_{\max}$. (b) The identifiability ratio, one point per observer, at $r = 2$ (open) and $r = 6$ (solid).

The ceiling. At the frozen eight-component configuration the estimator saturates above about eight components; at $k = 20$, under the twenty-component one, it tracks the rank to about ten and then falls behind (table 9). Over every configuration we swept the largest r it tracks follows neither the embedding dimension nor the record length and never exceeds about eleven (Appendix L). Most of it is the number of components the delay window resolves linearly: a linear participation ratio on the same covariance saturates along the same curve.

The zero-learning-rate test. The check of section 4 invalidated four of our six systems and the function-subspace variant besides: each returns its reported estimate with the parameters frozen, because the forcing keeps the residual moving while the parameters do not. It also rejects the instantaneous mini-batch loss of this system, otherwise among the more accurate observers.

6 CONDITIONS OF VALIDITY

6.1 Recovery in each regime

In figure 2(a) the fast quasiperiodic forcing follows the diagonal over the whole withheld range. Under mini-batch noise the estimate is flat in r and well above the truth, responding to $E_{\max}$ and to the length of the record instead of to the data. Noise at a fortieth of the forcing amplitude leaves the ordering nearly intact; it still moves the estimate at $r = 1$ from 1.1 to 11.3. The slow forcing changes behaviour partway along: it tracks the truth to about $r = 4$, then saturates.

Panel (b) plots the identifiability ratio on either side of that change. The regime labels alone would have

predicted the extreme cases, not the slow forcing, which passes at $r = 2$ and fails at $r = 6$. The ratio therefore locates the failure inside a single system, using nothing the ground truth supplied.

The Theiler exclusion is why the estimator errs most on a transient: the 16 it returns there is a divergent quantity evaluated at the exclusion the configuration asks for. Removing it returns the value maximum likelihood gives on a uniformly sampled curve, at every rank, seed and observer, and leaves the fast recurrent case unchanged (Appendix I.2).

Every validation system above is externally forced. Training is not. The evaluation therefore establishes a conditional result. Full-batch descent does reach the edge of stability (Cohen et al., 2021), where an unforced run might still recur, but oscillation is not recurrence and the diagnostics reject every run at the edge (Appendix K).

6.2 The delay lag

The delay window spans $(E_{\max} - 1)\tau$ samples. Below an appreciable fraction of the oscillation period the delay coordinates are nearly collinear and the torus never unfolds (Casdagli et al., 1991); varying τ alone on a fixed torus moves the estimate over more than an order of magnitude, through the truth only where the window covers a few tenths of a period (figure 6). The lag must be set from the oscillation period, which a constructed system supplies and a training log does not, so from the grokking logs we take changes and never a level (section 7).

6.3 Window-size limits

We distinguish the delay span from the local window. W is the number of samples pooled into one estimate; $S = (E_{\max} - 1)\tau$ is the span of one delay vector; P is the recurrent period; and N is the full record. Thus S/P selects the delay τ , whereas W selects locality. The implementation rejects $W < E_{\max}\tau + 20$; a valid estimate also needs independent returns and a window contained in one regime. The working choices are summarised in table 2; the complete sweeps are in Appendices D, H and L.

The reference band. The band used below is empirical, based on the accurate validation cases, and is not calibrated beyond them.

6.4 Nuisance factors and nulls

Controls in Appendix M show that observer gain can shift the estimate at fixed truth, whereas phase changes can occur at fixed roughness; a falling scalar estimate

Table 2: Working window sizes. The values are empirical choices for the corresponding analysis; W does not determine the delay span S .

analysis	local window W	interpretation
controlled recurrent systems	8000 samples	dimensional level, only if recurrence and identifiability diagnostics pass; choose τ so $S/P \simeq 0.2\text{--}0.4$ when P is known
training logs, coarse diagnostic	$\simeq 4000$ samples	diagnostic only; the level is not an active dimension
training logs, transition comparison	60 samples	change statistic only; resolves transitions of hundreds of steps
stored full-batch trajectory	600 published; $10^4\text{--}3 \times 10^4$ sweep	600 is the local measurement; longer windows reveal drift or a feature but do not separate runs

is therefore not sufficient evidence of a dimensional change.

The effective rank. Unequal phase amplitudes lower effective rank while the active dimension stays fixed; the estimate follows only the phases it resolves (figure 7).

7 APPLICATION TO GROKING

7.1 The diagnostics reject both settings

We run the frozen configuration on these logs, shortening the window and stride to fit (table 14, figure 8). Transformer estimates are unstable (four to thirteen components), while the perceptron and no-decay runs look transient: their trend-crossing ratios remain outside the valid band. The level is therefore not a dimension in either setting; the coarse window smears a transition of a few hundred steps across tens of thousands.

7.2 The trajectory simplifies at generalisation

A stored trajectory permits a measurement that a scalar log does not. We compress the parameters and probe logits by random projections (Appendix G.1). Near generalisation, detrended effective rank collapses in both spaces, reaches near unity in parameter space, and then re-expands (figure 3, table 16). The timing repeats across seeds with a fourfold spread in transition time and on a second task (figure 10); it is not explained by the later floor in displacement. Because effective

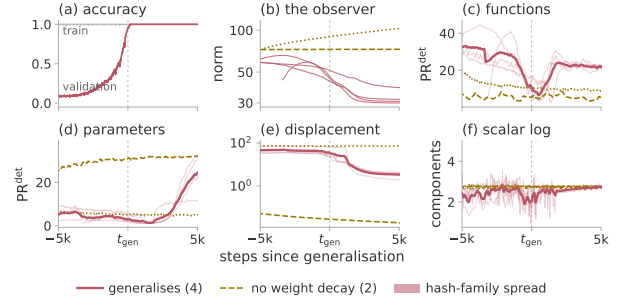


Figure 3: Simplification of the trajectory at generalisation. Three regularised seeds of modular addition and one S_5 composition against the same configurations without weight decay, each aligned on the generalisation step of the run it matches. Thick line, the mean of the four generalising runs; band, the hash families of Appendix G.1. (a) is the one run whose training curve is stored, (b) the observer for all six, (c)–(e) PR^{det} in each space and the displacement, over windows of 600 steps (300 for S_5), and (f) the estimate of section 7.3 on (b).

rank measures variation inside the window, this collapse alone need not imply a change in $d_{\text{act}}(R)$.

The runs without weight decay and matched pairs. No-decay runs memorise and stay at chance; one later dips as far as a generalising run (table 16). Timing and reversal separate the pairs; details are in Appendices F and H.1.

7.3 The log estimate falls where the trajectory does

A transition-scale window makes the comparison possible, but it need not lie in one regime, so its fall is a change statistic rather than a measurement of equation (1). We rerun the estimator on the same windows as the trajectory measurement, using a grid chosen without the outcome (Appendix H.3). On the parameter norm the estimate halves at generalisation in all four positive runs; 24/26 usable cells separate them from the two controls, and surrogate tests retain the effect (table 18 and Appendix H.3). The log therefore records the same event as the stored trajectory, but more weakly: on a training log, observer smoothness remains an alternative explanation.

8 CONCLUSION

In a deterministic recurrent system with a matched lag, a suitable scalar log counts active components to about one component, up to a ceiling near eight. Outside that regime, transience and mini-batch noise invalidate the

dimensional interpretation; the diagnostics reject both grokking settings. Their stored trajectory nevertheless shows a detrended-rank collapse and recovery near generalisation, with timing and reversal separating four positive runs from two controls.

Limitations. Validation is externally forced, whereas training is not; recurrence and a period-matched lag are required, and the estimator is ordinal above about eight components. The neighbour-scale power law is unchecked, while the grokking signal is affected by weight decay, observer smoothness, and the limited four-versus-two comparison; it does not reproduce in the full-batch setting (Appendices H.2 and M). Edge-of-stability evidence comes from one task (Appendix K).

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APPENDIX OUTLINE

These appendices are supplementary. The body stands without them; each holds the data, the construction or the audit behind a claim made there.

Appendix A states, without proofs, the four published results the estimator is assembled from: the delay embedding theorem of Takens (1981), its extension to fractal sets, the version for systems forced by an external input, and the maximum likelihood estimator of an intrinsic dimension. It closes with the limits a finite record imposes.

Appendix B records the procedure as run: the algorithm and its constants, the scalar observers of section 5.1, and the configurations at which the estimator was frozen, with the grids they were selected on.

Appendix C scores section 5 in full, observer by observer, against every simpler statistic we could compute on the same data. Three of those comparisons go against the estimator.

Appendix D gives the three experiments section 6 summarises, on the delay lag, the nuisance factors and unequal forcing amplitudes, and a fourth the body does not report: whether a real change in the active dimension is detected when one occurs.

Appendix E describes how the image-data system of section 5.1 is built in the three variants the paper scores, and measures how many independent directions each excites. The function-space variant falls short of its own requirement.

Appendix F gives the diagnostic values behind section 7.1 for every training log, and shows the label-

matched pairs beside the raw loss curves that limit any claim of a separation between them.

Appendix G covers the measurement of section 7.2: the random projection that makes storing a whole trajectory affordable, the checks that validate it, the spectral guarantee those checks do not supply, and the collapse run by run.

Appendix H makes one argument three times: a statistic computed over a window cannot localise anything shorter than that window. The three parts make it for the full-batch trajectory, for a run whose estimate declines although it never generalises, and for the shortened window of section 7.3, where that result is also qualified.

Appendix I audits the Theiler exclusion, the one setting that fixes the value returned on a transient and the one at which this paper departs from its own protocol. It measures the cost of that departure and varies the exclusion from zero upwards.

Appendix J gives every run its task, architecture, length and outcome, and states where our grokking settings depart from the published configurations they are taken from.

Appendix K tests whether gradient descent at the edge of stability is deterministic and recurrent at once, as the estimate requires, and how much of that survives a coarser log.

Appendix L separates the level the estimator returns at high rank from the highest rank at which it still tracks the truth.

Appendix M varies the number of phases at a fixed roughness and the roughness at a fixed number of phases. It also reports two findings that count against the estimator.

Appendix N states the compute used and the material to be released.

A THE RESULTS BEHIND THE ESTIMATOR

The results below are stated without proof and in the form the paper uses. Every condition they impose is an assumption about the run rather than a property we can verify from the log.

Throughout, f is the map advancing the system by one step, ϕ the scalar observation, and

$$\Phi_{f,\phi}(\mathbf{s}) = (\phi(\mathbf{s}), \phi(f(\mathbf{s})), \dots, \phi(f^{E-1}(\mathbf{s}))) \in \mathbb{R}^E \quad (4)$$

the delay map.

Theorem 1 (Delay embedding; Takens, 1981). *Let M*

be a compact manifold of dimension d and let $E > 2d$. For a generic pair (f, ϕ) with $f \in C^2(M, M)$ a diffeomorphism and $\phi \in C^2(M, \mathbb{R})$, the delay map equation (4) is an embedding of M into $\mathbb{R}^E$.

An embedding is one-to-one and an immersion, so distances in $\mathbb{R}^E$ between reconstructed points are a faithful, though distorted, image of distances between states. The set they fill is the object the neighbour statistic below measures (figure 4). Theorem 1 requires a manifold and the sets a training run occupies need not be one; the extension below is the version the paper relies on.

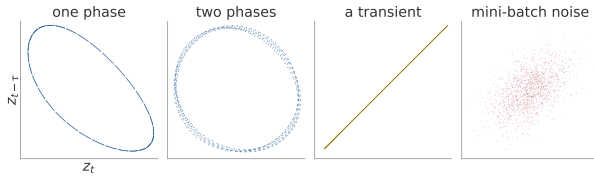


Figure 4: The set the neighbour search measures, in the plane of the first two delay coordinates. One window of the image-data system of section 5.1 under each excitation, reconstructed at the frozen configuration. A single phase closes an orbit, two fill a surface, a converging run never returns, and mini-batch noise fills the space it is given.

Theorem 2 (Fractal delay embedding prevalence; Sauer et al., 1991). *Let $A \subset \mathbb{R}^n$ be compact with box-counting dimension d , invariant under a smooth diffeomorphism f defined on an open neighbourhood of A , and let $E > 2d$ be an integer. Assume that for each $p \leq E$ the set of periodic points of f of period p has box-counting dimension less than $p/2$, and that the linearisation of f at each such orbit has distinct eigenvalues. Then for almost every smooth ϕ , in the sense of prevalence, the delay map equation (4) is one-to-one on A and an immersion on every compact subset of a smooth manifold contained in A .*

The dimension in the condition $E > 2d$ is the box-counting dimension of the occupied set $\mathcal{A}_R$, which need not be an integer, so the criterion applies to the sets a training run occupies and not only to tori. Prevalence, in turn, is a measure-theoretic notion of “almost every” on an infinite-dimensional space.

Mini-batch sampling makes the update depend on a draw from the data as well as on the state, so neither theorem above applies as stated.

Theorem 3 (Delay embedding for forced systems; Stark, 1999; Stark et al., 2003). *Let $\mathbf{s}_{t+1} = f(\mathbf{s}_t, w_t)$ with the forcing w_t generated by a measure-preserving system on a space Ω , either deterministically or as an independent draw selecting among finitely many maps.*

Under genericity conditions on f and ϕ analogous to those of theorem 1, and for E exceeding twice the dimension of a fibre, the map sending a state to its delay vector equation (4) is an embedding of each fibre M_ω , that is, of the set of states compatible with one realisation ω of the forcing.

The guarantee holds fibre by fibre, whereas the sample available in practice is one trajectory, which lies in a different fibre at every step. No invariant measure on a fibre is being sampled, so the neighbour statistic below estimates nothing.

Theorem 4 (Maximum likelihood intrinsic dimension; Levina and Bickel, 2004; MacKay and Ghahramani, 2005). *Let points be drawn from a smooth density supported on a d -dimensional manifold embedded in $\mathbb{R}^E$. Model the points falling within a small ball about $\mathbf{y}_i$ as a homogeneous Poisson process in the d -dimensional radial variable, with the density constant over the ball. Writing $r_j^{(i)}$ for the distance from $\mathbf{y}_i$ to its j -th nearest neighbour, the maximum likelihood estimate of d from the first m neighbours of that point is*

$$\hat{d}_m(\mathbf{y}_i)^{-1} = \frac{1}{m-1} \sum_{j=1}^{m-1} \log \frac{r_m^{(i)}}{r_j^{(i)}}. \quad (5)$$

Under this model the reciprocal is unbiased and the estimate itself is not.

The estimator is therefore a local statistic: it reports the exponent over the range of radii the m nearest neighbours span rather than the limit equation (1) takes. For $0 < \varepsilon_1 < \varepsilon_2$ that exponent at a point $\mathbf{c}$ is

$$d_R(\mathbf{c}; \varepsilon_1, \varepsilon_2) = \frac{\log \mu_R(B(\mathbf{c}, \varepsilon_2)) - \log \mu_R(B(\mathbf{c}, \varepsilon_1))}{\log \varepsilon_2 - \log \varepsilon_1}. \quad (6)$$

Its median over $\mathbf{c} \sim \mu_R$ is the finite-scale active dimension $d_{\text{act}}(R; \varepsilon_1, \varepsilon_2)$ of section 3.2, which equation (3) estimates by pooling over the N points of the record. It agrees with $d_{\text{act}}(R)$ when the same power of ε continues below ε_1 . A phase of smaller amplitude than ε_1 is invisible to it however long the record. A record of N points resolves a dimension of at most about $2 \log_{10} N$ under the argument of Eckmann and Ruelle (1992), a bound appendix L finds does not explain the ceiling we measure.

B THE ESTIMATOR, ITS CONFIGURATION AND THE OBSERVERS

B.1 The estimator

Algorithm 1 is the estimator as it was run, with the departures from equation (3) that affect the values reported.

Algorithm 1 Pooled Levina–Bickel intrinsic dimension of a delay reconstruction

Require: series $x_{1:n}$; lag τ ; embedding dimension E ; neighbours m ; Theiler exclusion W_T ; dither σ

- 1: $z_t \leftarrow (x_t - \text{mean } x) / \text{std } x + \sigma \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, 1) \quad \triangleright$ breaks exact ties
- 2: $\mathbf{y}_t \leftarrow (z_t, z_{t-\tau}, \dots, z_{t-(E-1)\tau})$ for $t = (E-1)\tau + 1, \dots, n$
- 3: $S \leftarrow 0; N \leftarrow 0; n_{\text{dist}} \leftarrow 0; n_{\text{ratio}} \leftarrow 0$
- 4: **for each** $\mathbf{y}_i$ **do**
- 5: $r_1^{(i)} \leq \dots \leq r_m^{(i)} \leftarrow$ distances to the m nearest $\mathbf{y}_j$ with $|i - j| > W_T$
- 6: $r_j^{(i)} \leftarrow \max(r_j^{(i)}, \varepsilon_{\text{dist}})$ for each j ; count those clipped into n_{dist}
- 7: $S_i \leftarrow \max(\sum_{j=1}^{m-1} \log(r_m^{(i)} / r_j^{(i)}), \varepsilon_{\text{ratio}})$; count clips into n_{ratio}
- 8: $S \leftarrow S + S_i; \quad N \leftarrow N + 1$
- 9: **end for**
- 10: mark the window degenerate if n_{dist} or n_{ratio} exceeds 1 % of its denominator
- 11: **return** $(N(m-1) - 1) / S$

The returned value is $(N(m-1) - 1) / S$, the exactly unbiased pooled form, which differs from equation (3) negligibly at the sizes used here. A window marked degenerate is returned rather than discarded. No published cell of table 14 contains a marked window. The returned value is never clamped to E , so a divergent estimate is reported as divergent.

The Theiler exclusion is $W_T = \max((E-1)\tau, t_{\text{acf}})$, where t_{acf} is the first lag at which the autocorrelation falls below $1/e$, capped at 150 samples; appendix I.1 measures the effect of that cap. The floors $\varepsilon_{\text{dist}}$, $\varepsilon_{\text{ratio}}$ and the dither σ are 10^{-8} , 10^{-5} and 10^{-9} .

B.2 The observers

Table 3 defines the twelve observers of section 5.1.

Table 3: The scalar observers. $\mathbf{c}$ is the coordinate in the k -dimensional subspace and $\boldsymbol{\theta}_0$ the frozen head at initialisation; $\mathbf{L}$ is the matrix of probe logits, where the probe is a fixed held-out set drawn once before training; $\mathbf{g} = \partial \ell / \partial \mathbf{c}$. The projection directions $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{a}$, and the rotation $\mathbf{R}$, are drawn once from a fixed seed and never redrawn.

name	family	definition
instantaneous loss	loss	the loss on the current minibatch
full-batch loss	loss	the loss on the whole training set
probe loss	loss	cross-entropy on the probe set
parameter norm	norm	$(\ \boldsymbol{\theta}_0\ ^2 + \ \mathbf{c}\ ^2)^{1/2}$
subspace norm	norm	$\ \mathbf{c}\ _2$
function-space norm	norm	$\ \mathbf{L}\ _F$
gradient norm	gradient	$\ \mathbf{g}\ _2$
gradient projection	gradient	$\langle \mathbf{u}, \mathbf{g} \rangle$
fixed parameter projection	projection	$\langle \mathbf{v}, \mathbf{R}\mathbf{c} \rangle$
function-space projection	function	$\langle \mathbf{a}, \text{vec } \mathbf{L} \rangle$
margin	function	mean probe logit gap, true class less best other
probe accuracy	function	accuracy on the probe set

B.3 Frozen configurations

One frozen configuration serves each range of the active dimension, each selected once on data withheld from every later experiment (table 4).

Table 4: The frozen configurations and their selection. Only the swept parameters were varied; the remainder were fixed before any estimate was computed. Settings common to both configurations are not repeated: $m = 20$ neighbours, a Theiler exclusion of at least the embedding span (appendix B.1). The neighbour count and the exclusion were swept in the eight-component grid, in which the two Theiler rules return bit-identical values, so the selection between them is arbitrary.

	eight-component	twenty-component
E_{max}	20	40
τ	4	16
window / stride	8 000 / 2 000	8 000 / 4 000
swept	$E_{\text{max}}, \tau, m, \text{Theiler}, \text{window}$	E_{max}, τ
grid size	48	4
selection seeds	90, 91, 92	0, 1, 2
selection ranks	2, 4, 6	2, 6, 10, 14, 18
withheld seeds	0–3	none
withheld ranks	1, 3, 5, 8	the other fifteen
selection error, best	0.17	2.47
selection error, worst	2.03	3.35

The selection objective in each case is the error of the median estimate against the median measured effective rank, over the four observers reserved for selection. The eight-component objective adds a penalty on the rank correlation and on the spread across seeds.

Each configuration lies at a grid boundary, at the top of every axis it was swept over, so a wider grid could move the level.

The selection error spans a factor of twelve on identical data (table 4), so the level is as much a choice as a measurement. Only the withheld numbers of section 5.1

demonstrate recovery.

The stride varies between experiments. Table 5 states the window geometry each experiment uses; none of them overrides a setting of the estimator. The constructed systems produced the recovery of section 5.1 at a stride of their own, so “at the frozen configuration” is true there of the estimator and not of the stride.

Table 5: The window and stride of each experiment, with their departures from the frozen configuration of table 4. Lengths are in samples of the log; the second stride applies to the longer records.

experiment	window	stride
frozen configuration	8 000	2 000
constructed systems (section 5)	8 000	3 000, or 3 666
training logs (section 7)	a third of the record	1 000

C PER-OBSERVER RESULTS AND ALTERNATIVE STATISTICS

Table 6: Recovery on each system, with the requirements of section 4 that each fails. Systems are ordered by increasing realism. The unit of analysis is one (rank, seed, observer) run, each side of the error being the median over that run’s sliding windows; ρ is over ranks after a median over seeds. The oscillating matrix has no optimiser, so the zero-learning-rate check does not apply to it. The function-subspace row fails constructed truth in half its withheld cells, which reach about $0.87r$ against the $0.9r$ the check asks.

system	truth	range	MAE	ρ	requirements failed
oscillating matrix	constructed	1–8	0.27	1.00	<i>not applicable</i>
online linear regression	constructed	1–20	0.79	0.99	frozen configuration, zero learning rate
logistic regression	constructed	1–20	0.81	1.00	frozen configuration, zero learning rate
frozen nonlinear decoder	constructed	1–20	1.58	0.94	zero learning rate
perceptron in a k -subspace	constructed	1–20	0.92	0.99	zero learning rate
image data, parameter subspace	measured	1–8	0.90	1.00	none
image data, parameter subspace	measured	1–20	1.75	1.00	withheld ranks
image data, function subspace	measured	1–8	0.28	1.00	all of them

All numbers in this appendix are computed on the withheld ranks $r \in \{1, 3, 5, 8\}$ and the withheld seeds of section 5.1, in the deterministic recurrent regime, scored against the measured effective rank rather than against r . Table 7 reconciles the three errors this paper quotes for that one measurement.

Table 7: Three errors for one measurement, and where each is quoted. Each row is the same eight-component recovery of section 5.1, on withheld ranks and withheld seeds in the deterministic recurrent regime, summarised in a different order. The order is stated wherever one of these figures appears.

aggregation	reported in	error
mean over (rank, seed, observer) runs	section 5.1, table 6	0.90
median over observers of each observer’s error	table 8	0.48
median over seeds and observers at each rank, then scored	table 10	0.382

Table 8: The accuracy of each scalar observer, against the smoothness that might explain it. One row per observer, except that the parameter norm and the subspace norm share one, their errors agreeing to two decimals. The slope is that of the estimate on the truth. The rank correlation is 1.00 for every observer, so it has no column. The last column records whether the roughness orders the withheld ranks perfectly.

observer	family	MAE	slope	roughness orders rank
probe loss	loss	0.60	0.82	no
function-space norm	norm	0.39	0.96	no
margin	function	0.46	0.83	no
parameter, subspace norm	norm	0.41	1.04	no
fixed parameter projection	projection	0.34	0.83	no
gradient projection	gradient	0.70	0.72	no
function-space projection	function	0.51	0.78	no
gradient norm	gradient	1.81	1.58	no
full-batch loss	loss	1.87	1.51	no
instantaneous loss	loss	0.61	0.73	no
probe accuracy	function	degenerate in every window		

Two observers are excluded from every aggregate. Probe accuracy is quantised, which collapses the delay vectors onto one another. The instantaneous mini-batch loss fails the zero-learning-rate check: frozen, it still climbs by more than five components across the withheld ranks and returns the estimate it returns when trained. The other eleven observers are exactly constant at zero learning rate, so no estimate is defined for them.

Nothing in a training log says which observer is the accurate one, so a reader can act only on the typical observer and not on the best row of table 8.

The highest rank correlation the roughness reaches over the four withheld ranks is 0.8, so smoothness alone accounts for no observer’s agreement with the truth (figure 5).

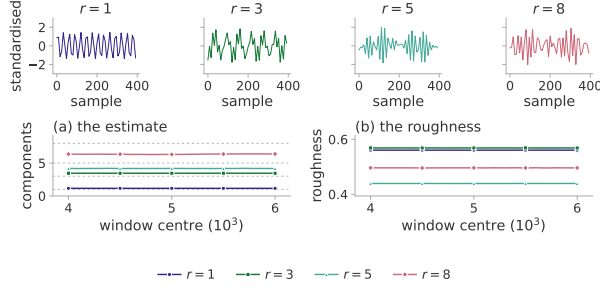


Figure 5: Four logs that look alike, separated by the estimator. The image-data system of section 5.1 at four ranks and a seed the configuration was not selected on, through the parameter norm. Above, four hundred samples of each log. (a) The estimate over sliding windows, the measured effective rank dotted. (b) The roughness of the same windows.

Table 9: At $k = 20$ the rank is tracked to about ten and then lost. Image data, median over three seeds and over the eight observers recorded there, which reports the typical observer rather than the best one; the truth is the measured effective rank. The last row is the spread of MG across seeds at fixed rank. Exploratory: this grid was scored with a smaller Theiler exclusion than Algorithm 1 asks (appendix I.1).

true r	1	2	4	6	8	10	12	14	17	20
MG	1.10	2.05	4.51	6.51	7.45	9.92	10.23	11.40	13.90	15.44
PR _{delay}	2.00	2.26	3.79	5.42	7.90	9.45	9.43	11.25	11.43	13.42
spectral PR	1.09	1.27	2.13	2.89	4.10	4.60	5.51	6.22	6.81	7.48
spread across seeds	0.87	0.51	1.35	1.81	2.64	3.59	2.81	2.07	3.41	4.24

Table 10: Every simpler statistic we could compute on the same data, scored beside the estimator on the image-data system of section 5.1. The two ranges are not aggregated alike: at $r \leq 8$ each statistic is scored on the withheld ranks alone, at $r \leq 20$ on all twenty, the selection ranks included, so only the comparison down a column is out of sample. Statistics marked † need a neighbour search; the rest are one-line computations on the series or its spectrum.

statistic	$r \leq 8$		$r \leq 20$	
	MAE	ρ	MAE	ρ
MG †	0.382	1.00	1.75	0.997
LB †	0.40	1.00	1.52	0.997
TwoNN †	2.13	0.80	2.79	0.880
PR _{delay}	0.91	1.00	2.48	0.985
spectral PR, 256 bands	1.25	1.00	5.83	0.998
spectral PR, 1024 bands	1.08	1.00	4.78	0.979
spectral PR, native resolution	0.69	1.00	2.95	0.976
roughness	3.65	0.20	9.89	-0.379

Three qualifications attach to table 10. LB, the uncorrected per-point average, is more accurate than MG at

$r \leq 20$: the correction of section 3.3 removes a positive bias, which on a saturating estimate was partly cancelling the saturation. The spectral participation ratio at 256 bands orders the ranks at $r \leq 20$ more faithfully than MG, at three times the error. The linear delay participation ratio needs no neighbour search at all, yet MG beats it by only a factor of 2.4 at $r \leq 8$.

D CONDITIONS OF VALIDITY IN DETAIL

The delay lag. Figure 6 varies the lag τ alone. Below a tenth of a period every rank above one returns about 2.5 and the ordering is lost; between three tenths and a half of a period the estimate follows the rank; above one period it diverges.

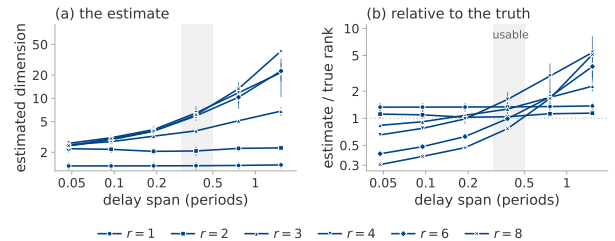


Figure 6: The delay lag, varied on one system: a fixed torus, two seeds, six lags spanning a twentieth of a period to one and a half periods. Marker shape is the true rank, the marker the median of the seeds and the bar their range. (a) The estimate. (b) The same divided by the true rank, so one reference line replaces six; the shaded column is three tenths to a half of a period.

Nuisance factors. The rotated-coordinate row of table 11 is scored on the fixed parameter projection alone, since the other eleven observers are invariant to a rotation by construction; it measures a property of the observer rather than of the estimator.

Table 11: Shift in the estimate produced by factors that leave the active dimension unchanged, median over five seeds and twelve observers. The first row is the seed-to-seed variability of an unchanged configuration and sets the scale for the others. The last column is a percentage of a real four-component change, which the change-detection experiment below observes as a fall of 3.96. A constant rescaling of the observer moves the estimate by 0.000 and is omitted.

factor	$ \Delta \text{MG} $	as % of a real four-component change
none (baseline)	0.39	10
learning rate halved	0.37	9
forcing band up one octave	0.50	13
forcing band down one octave	0.78	20
coordinates rotated	0.28	7
observer gain ramped $10\times$	1.53	38
state amplitude ramped $4\times$	1.57	39

Detection of a change. We run three consecutive segments $r_{\text{high}} \rightarrow r_{\text{low}} \rightarrow r_{\text{high}}$, re-measuring the ground truth on each. Over the three schedules and twelve observers, a deterministic recurrent fall of 4.00 components is observed as 3.96, is detected both downward and upward on over nine tenths of the runs, and recovers to within 0.05 of the level before it. Under stochastic excitation the same true change is observed as 0.09. Table 12 gives each schedule separately, on the parameter norm.

The rule declares a change when the estimate falls below the median of the pre-switch segment by three times the window-to-window scatter of that segment, for two consecutive windows.

Table 12: A change of the active dimension is recovered at both levels and after it reverses. Three schedules $r_{\text{high}} \rightarrow r_{\text{low}} \rightarrow r_{\text{high}}$, on the parameter norm. At $6 \rightarrow 2 \rightarrow 6$ the high level is over-reported by a component.

schedule	high	low	high again
$4 \rightarrow 1 \rightarrow 4$	4.06	0.84	4.06
$6 \rightarrow 2 \rightarrow 6$	7.00	2.17	6.95
$8 \rightarrow 3 \rightarrow 8$	8.32	3.34	8.28

Unequal forcing amplitudes. Figure 7 gives the measurement of section 6.4 in full, observed through the squared Frobenius norm and estimated with the frozen eight-component configuration.

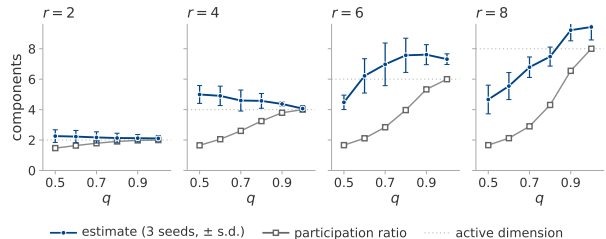


Figure 7: The estimate follows the active dimension, not the effective rank. Forcing phase l at amplitude q^l holds the active dimension at r while the effective rank falls with q . The system of section 5, three seeds, one panel per r ; the dotted line is r . Filled circles, the estimate, median over seeds, with standard deviation as bars; open squares, the measured effective rank, which matches its closed form to two decimals and so has no spread.

The estimate is flat in q at $r = 2$ and $r = 4$ while the effective rank declines, as intended, and declines itself at $r = 6$ and $r = 8$. The amplitudes show why: at $q = 0.5$ and $r = 8$ the weakest phase is forced at 0.008 of the strongest, far below the neighbourhood scale of the estimator, so the torus it resolves really does have fewer dimensions than the one that was built. The estimator therefore reports the number of phases it can resolve. That is neither r nor the effective rank; the three coincide only when the amplitudes are equal.

E SYSTEM CONSTRUCTIONS

The image-data system of section 5.1. The data is partitioned into twelve fixed groups, the same partition at every rank. The loss weights of r of those groups are modulated by r rationally independent sinusoids filling one octave. Modulating group j tilts the gradient along a direction ϕ_j . These directions are neither orthogonal nor of equal effect, since random data groups have correlated gradients, so a mixing matrix $\text{pinv}(\Phi)Q$ with Q an orthonormal basis of $\text{range}(\Phi)$ orthogonalises them. The per-phase scalar response gain of the linearised update is then divided out.

Table 13 measures how many independent directions each construction excites. The transient and the plain mini-batch rows report an outcome rather than a setting: a converging trajectory is a curve whatever r is and the rank of mini-batch noise is fixed by the per-example gradient covariance of the data, so projecting it onto r directions recovers only part of the intended rank and the estimates in that regime are scored against the measured value.

Table 13: The independent directions each construction excites, against the number intended. Measured trajectory participation ratio by excitation regime and nominal rank, median over the four seeds, for the ten available directions of section 5.1. Every estimate on this system is scored against these values, never against r . On the **qp** rows that carry the recovery claim of section 5.1 the two agree to three decimals and the choice moves no error by more than 0.001; it matters on the rows where the construction fell short.

regime	$r = 1$	2	4	6	8	excitation
qp	1.00	2.00	4.00	6.00	8.00	r sinusoids, fast forcing
qp_slow	1.00	2.00	4.00	5.99	7.98	the same at a training-log timescale
qp_nopre	1.00	2.00	3.73	5.98	7.77	qp without the preconditioner
mixed	1.00	2.00	4.00	6.00	7.99	qp plus noise in the same directions
noise	1.00	2.00	4.00	5.99	7.99	injected rank- r gradient noise
noise_nopre	1.09	1.98	3.71	5.24	6.75	the same without the preconditioner
batch_proj	1.00	1.74	3.08	4.19	5.38	real mini-batch noise projected to rank r
batch	6.58	6.58	6.56	6.60	6.56	plain mini-batch descent
gd	1.04	1.05	1.04	1.04	1.06	a decaying transient
qp, $\eta = 0$	1.00	1.00	1.00	1.00	1.00	the zero-learning-rate check

The functional rank of the backbone is 10 at every rank, seed and regime, a property of the frozen backbone rather than of the excitation.

The function-space construction. A local adapter is built around a trained network of widths 64, 16 and 10. Its function-space Jacobian is QR-whitened, so that the first r columns have rank r at equal local scale. A quasiperiodic teacher excites every adapter coordinate and the student tracks it. The functional, trajectory-covariance and update-covariance ranks are measured rather than assumed. Eight of the sixteen withheld cells fall short of the $0.9r$ effective rank the construction requires, at $0.87r$ in the median. Table 6 therefore marks this variant as failing every requirement.

The system at twenty available directions. The same construction at $k = 20$ reproduces the nominal rank over $r = 1, \dots, 20$ in the recurrent regime; every other regime behaves as it does at $k = 10$.

F DIAGNOSTICS AND FIGURES FOR THE GROKING APPLICATION

Table 14 gives the diagnostics behind section 7.1.

Table 14: The estimate is a dimension in neither grokking setting. The estimator and its diagnostics on training logs at $E_{\max} = 20$: the regularised stochastic setting through the parameter norm, the unregularised full-batch setting through the training loss. Every cell is a median over the run’s sliding windows. Seven of the seventeen logs figure 8 plots are shown here. A “no” is a run that had not generalised when training ended, which is a censored observation.

run	generalises	MG	ρ_{ident}	roughness	PR _{delay}	crossings
transformer, full data	yes	13.32	1.53	0.097	1.03	204
transformer, reduced data, s0	no	4.70	1.59	0.067	1.50	161
transformer, reduced data, s2	no	4.42	1.55	0.139	4.04	153
transformer, no weight decay	no	15.60	1.76	< 0.001	1.00	2
perceptron, $n + m$	yes	15.08	1.42	0.001	1.00	2
perceptron, $n \cdot m$	yes	14.97	1.39	0.001	1.00	2
perceptron, $n^3 + nm^2 + m$	no	15.41	1.51	< 0.001	1.00	2

Figure 8 places every training log of section 7 in the plane of the identifiability ratio and the trend-crossing count, against the region in which every accurate case of section 6 fell. No run lies in that region: every one is unstable in the embedding dimension, at ratios well above the 1.10 the region allows. On the other axis the transformer and perceptron runs fail in opposite directions, since the perceptron series are monotone.

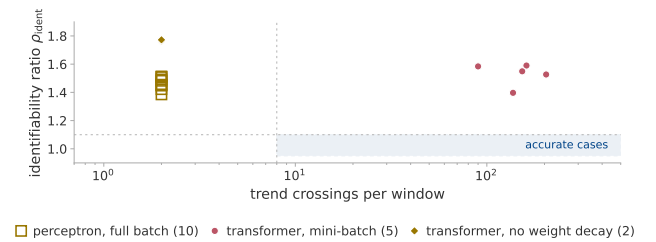


Figure 8: No grokking log lies where the estimate was accurate. Every log of section 7 in the plane of the identifiability ratio and the trend-crossing count, the transformer runs observed through the parameter norm and the perceptron runs through the training loss. Colour is the regime of section 3.4, marker shape the setting. The shaded rectangle is where every accurate case of section 6 fell. It is a record of those cases, not a test.

Figure 9 shows the separation of section 7.2 and the mechanism behind it: the medians separate every pair, though the ranges overlap.

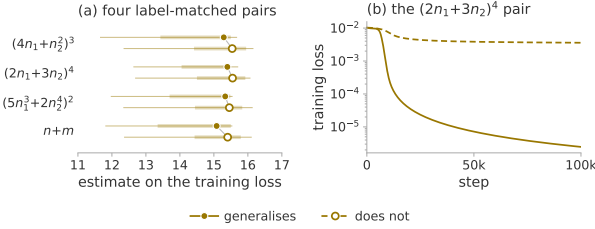


Figure 9: The estimate tells apart two targets that fit the training set alike, only one of them representable by the architecture. (a) The four pairs of section 7.2, on the training loss. Marker, median over the sliding windows; thick bar, interquartile range; thin line, full range. All runs are transient, hence one colour; fill and dash mark generalisation. (b) One pair in full: the representable target drives the loss orders of magnitude lower.

G THE DIRECT TRAJECTORY MEASUREMENT

G.1 The trajectory sketch

The direct measurement of section 7.2 needs the trajectory itself: the whole parameter vector at each of two to three thousand logged steps. Every logged step is therefore compressed to $\mathbb{R}^{1024}$ by a CountSketch (Charikar et al., 2002; Weinberger et al., 2009), a sparse random projection drawn twice from two independent hash families so that the two copies can be compared. Two spaces are recorded: the parameters, and the probe logits after centring each example over classes and normalising to unit length, without which a signal taken from the logits could not be told from a measurement of the weight decay its scale is coupled to.

We do not verify a spectral guarantee for the compression. The two hash families supply a direct estimate of the error instead: every reported value is the mean of the two projections, with the disagreement between them recorded beside it. That disagreement is a per cent of the value at the median and under nine per cent at its worst, smaller than every effect section 7.2 reports.

The sketch is accurate on synthetic trajectories of known rank. It agrees to within 0.11 components throughout (table 15). Nothing above rank ten is tested, however, whereas the plateaus in figure 3 are higher.

It agrees with exact arithmetic on a real run. On the perceptron architecture the participation ratio computed on all the raw parameters agrees with the sketched value to three decimals.

Recording the trajectory does not perturb training. The

log is required to be bit-identical with and without the recording, over four hundred steps at reduced probe size. One global random stream supplies the split, the initialisation and the mini-batch order in turn, so a single stray draw changes the initial weights and can destroy grokking.

Table 15: The cost of the compression. The sketch against the same statistic computed without it, on synthetic trajectories of 60 samples and known rank in 145 500 dimensions. The difference between the first two columns is the cost of a sixty-sample participation ratio and is not attributable to the sketch.

true rank	PR, uncompressed	PR, sketched	sketch – uncompressed
1	1.00	1.00	0.00
2	1.99	1.99	0.00
5	4.56	4.54	-0.02
10	8.44	8.33	-0.11

Windows are sixty logged rows, each labelled by its centre, so a feature at a given step is located to within half a window. The statistic reported per window is PR^{det} , the detrended effective rank of section 3.2. The total displacement is recorded beside it, since a rank can fall simply because the trajectory has stopped moving.

G.2 The effective-rank collapse, run by run

The plateau of table 16 is the median over the 4000 steps before the generalisation step, the dip the minimum within ± 4000 steps of it, and “at” the position of that minimum relative to it. The two runs without weight decay have no generalisation step, so each is measured over the window of the run it matches. In parameter space the post-dip maximum exceeds the plateau by an order of magnitude in the modular runs. In function space it returns to about the plateau and exceeds it in two of the four runs.

The same measurement rules out a collapse at one fixed absolute step. The four generalisation steps span 9980 steps and the minimum is searched over 8000, so no single step lies inside the search window of all four runs; every minimum in fact falls within 1513 steps of its own run’s transition, in both spaces (figure 10).

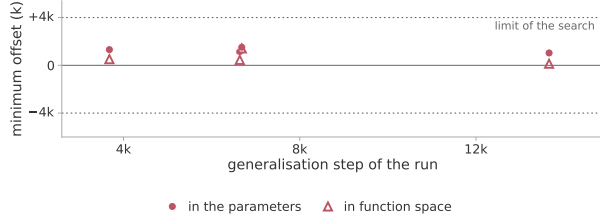


Figure 10: The collapse follows each run’s own transition. The minimum of the detrended participation ratio in each of the four generalising runs, against that run’s generalisation step. The two S_5 and modular runs near 6.7k nearly coincide. Dotted, the ± 4000 steps searched around each transition.

Table 16: Plateau, dip and recovery per run, in function space and in parameter space. Depth is plateau divided by dip. The post-dip maximum is the largest value reached afterwards: the sense of “re-expands” in section 7.2. The two rows without weight decay in each block are measured in the aligned window; their deepest fall anywhere in their own run is 3.00 and 7.03 in function space and 1.66 and 2.39 in parameter space, the comparison section 7.2 also reports.

space	run	plateau	dip	depth	at	post-dip max
function	mod_wd1	24.24	3.80	6.38	135	23.99
	mod_wd1_s43	33.02	3.86	8.55	455	27.80
	mod_wd1_s44	27.16	3.40	7.99	515	30.82
	s5_wd1	21.04	2.62	8.03	1413	23.53
	mod_wd0	5.50	3.43	1.60	—	—
	s5_wd0	11.87	9.09	1.31	—	—
parameter	mod_wd1	3.01	1.09	2.76	1035	36.12
	mod_wd1_s43	2.57	1.15	2.24	1155	37.34
	mod_wd1_s44	2.71	1.13	2.39	1315	37.44
	s5_wd1	7.14	1.39	5.14	1513	12.46
	mod_wd0	29.93	27.70	1.08	—	—
	s5_wd0	5.57	4.49	1.24	—	—

H THE RESOLUTION OF A WINDOWED STATISTIC

H.1 The full-batch setting

Repeating the direct measurement on the perceptron setting gives a limit of the measurement rather than a negative finding. Under full-batch gradient descent every window of every run has a participation ratio between 1.00 and 1.36 on the statistics the comparison uses.

A deterministic update makes the trajectory a smooth curve. Over a 600-step window such a curve is straight to within the measurement precision, so the statistic reports the curvature of the trajectory at the scale the window sets, where there is almost none. The statistic has no range in which a collapse could appear.

A range of window lengths settles the question (ta-

ble 17). Two runs of 150 000 steps, a generalising one and its label-matched partner, are measured over windows spanning 600 to 118 000 optimiser steps, each holding the same sixty samples spread further apart as the window lengthens, so that the bound on the statistic and its noise floor stay where the published measurement put them. The statistic does leave its floor, at a scale of about 10^4 steps, more than an order of magnitude above the published window. The absence of a feature at the shorter window was therefore a limit of resolution rather than a fact about the run.

The feature that appears there separates nothing. One localised maximum appears, of the same size in both runs (table 17); function space agrees, with more range and no more separation. Nor can a window this long place that maximum to better than the interval between memorisation and generalisation. Beyond 30 000 steps the ordering even reverses and the non-generalising run moves further, because the window is then a large fraction of a run whose weight norm is still climbing, so that the statistic tracks that drift. The full-batch trajectory therefore carries no signature of generalisation in its effective rank at any window length between 600 and 118 000 steps: the collapse of section 7.2 belongs to the regularised stochastic setting.

Table 17: The full-batch statistic leaves its floor only at the longest windows. Detrended position participation ratio in parameter space against window length, full-batch gradient descent at $p = 97$, over all placements of the window, sixty samples per window at every length.

window (steps)	generalises		control, never generalises	
	median	max	median	max
600	1.002	1.035	1.001	1.011
2 950	1.066	1.258	1.003	1.087
5 900	1.027	1.534	1.005	1.360
11 800	1.010	1.951	1.020	1.826
29 500	1.006	2.047	1.112	1.735
59 000	1.025	1.551	1.345	1.912
118 000	1.231	1.626	2.009	2.271

Introducing mini-batches confirms this. The same statistic then ranges over more than a decade while the generalisation step moves by a single logged step. Any claim based on a trajectory participation ratio must therefore be controlled for batch size and noise scale before it is attributed to learning.

H.2 Declines of the estimate that are not generalisation

A decline of the estimate is not specific to generalisation. In a long run at $p = 211$ with no weight decay, which memorises early and ends at chance validation

accuracy, a decline is detected tens of thousands of steps after memorisation. The ramped-gain experiment of section 6.4 supplies a mechanism for it that needs no dimension at all: the parameter norm of this run roughly triples, acting on the estimate through the observer. No trajectory was stored for that run, so nothing here says whether it simplified; at the frozen window and in these settings the estimate on a log is in any case neither a dimension nor an effective rank.

Section 7.3 does not make the decline specific either. The estimate on a log does decline where the trajectory does, in four runs and in neither of the other two, but at a window some seventy times shorter than the one used here, under a configuration chosen for that window.

H.3 Resolution of a windowed estimate on a training log

The change section 7.1 leaves open is testable on the seven extended runs. The frozen configuration takes a window of a third of each record, advanced by 10 000 steps at a time. That gives nine windows per run and localises a feature only to $\pm 20\,000$ steps. The runs therefore cannot be aligned on their generalisation steps as figure 3 aligns them: no two share an offset, so no mean over runs can be formed.

The one run with a window centre near its generalisation step *rises* by 2.93 components over the following stride and by 2.71 over the one after. Those are the only two of the fifty-six stride-to-stride changes to exceed the 1.53-component floor a ramped gain alone produces (section 6.4). The other two generalising runs stay inside it throughout, as do all four that never generalise (figure 12). A rise is not the fall section 7.3 reports, and at $\pm 19\,995$ steps neither could be assigned to the transition.

As in appendix H.1, the statistic has no range in which a feature confined to the transition could appear. Testing the claim requires a window set in units of the width of the transition, which for these runs is hundreds of steps.

The matched-window re-run. Section 7.3 performs that re-run on the six runs of figure 3 rather than the seven of this appendix: the question needs a paired comparison and only those six had their trajectory stored. Each window is 60 samples, centred on the midpoint of a window of section 7.2. The significance of the fall is assessed against the surrogates of section 7.3 (Theiler et al., 1992; Schreiber and Schmitz, 1996).

The frozen configuration does not fit. At 60 samples its delay span of 76 exceeds the window. The grid crosses $E_{\max} \in \{4, 6, 10\}$, $\tau \in \{1, 2, 4\}$, $m \in \{5, 20\}$ and both Theiler rules, giving thirty-six cells, of which twenty-six

are long enough to return a value. Twenty-four of those separate the four generalising runs from the other two. The headline cell is named by the rule of section 7.3, which does not choose the best cell: a neighbouring cell separates the runs more sharply.

The diagnostics do not survive the shortening. The identifiability ratio needs an embedding of $2E_{\max}$, so at the headline cell the Theiler exclusion leaves too few usable points to compute one. Where the ratio can be computed at all, the two Theiler rules disagree about whether it passes; it settles nothing.

Linear detrending removes the effect. Repeating every cell on a linearly detrended window leaves none that separates the generalising runs from the rest. Detrending a sketch and detrending a scalar are not comparable: on a 1024-dimensional sketch a per-coordinate linear detrend removes one direction out of many, whereas on a scalar series it removes most of the series.

The observer matters. On the training loss rather than the parameter norm, twelve of the twenty-six cells separate them, though the surrogate comparison passes in the S_5 run only.

The two runs without weight decay have almost no dynamic range. At this window the baseline belongs to the configuration rather than to the run: on a standardised straight line equation (3) returns about 2.77, close to the pre-transition level of every run. The parameter-norm log of the modular run is straight to under a per cent of its own spread, so its estimate spans three thousandths of a component against a range three orders of magnitude larger in the generalising runs; the S_5 run is a weaker case of the same thing. Their D and quantile columns therefore record an observer that offered the estimator no local structure to respond to; neither run is a negative result. The four generalising runs are untouched, and so is the surrogate comparison, in which each run is measured only against surrogates of itself (figure 11).

Table 18: The fall of the log estimate at generalisation. The matched-window re-run of section 7.3 on the parameter-norm log, at the headline cell $E_{\max} = 10$, $\tau = 1$, $m = 20$, embedding-span Theiler. D is the median estimate over the three thousand steps before the transition divided by the minimum over the window that brackets it; the quantile is the rank of D among the same statistic at every window centre after memorisation; the floor is where that minimum falls, relative to t_{gen} . The p columns are the fraction of surrogates, at three smoothing lengths, whose D is at least the observed one.

run	D	quantile	floor	p_{101}	p_{201}	p_{401}
modular, seed 42	2.96	0.99	-65	0.025	0.025	0.025
modular, seed 43	1.98	0.93	255	0.050	0.050	0.025
modular, seed 44	2.14	0.92	315	0.050	0.075	0.025
S_5 composition	2.12	0.89	1313	0.050	0.050	0.050
modular, no decay	1.00	0.46	-865	0.950	0.950	0.925
S_5 , no decay	1.04	0.38	1613	0.975	0.975	1.000

Put through the same comparison, the linear participation ratio responds less than the estimator and the roughness, as section 7.3 reports, far more.

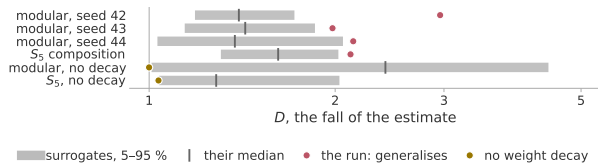


Figure 11: Each run’s fall against the null distribution built from that run alone. D is as in table 18, on the parameter-norm log at the headline cell. Band, the fifth to ninety-fifth percentile of the pooled surrogate draws, thirty-nine at each of five seeds; tick, their median; circle, the run. The runs without weight decay are measured in the window of the run each matches.

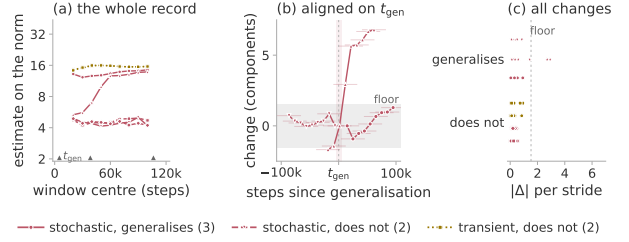


Figure 12: The frozen window localises a feature only to $\pm 20\,000$ steps. The estimate on the parameter norm for the seven extended runs, under the frozen configuration of section 7.1, plotted at the window centres. Colour is the regime of section 3.4. (a) Every window, with the generalisation steps marked on the axis. (b) The generalising runs aligned on their own generalisation step. (c) All the changes from one stride to the next, one sub-row per run. The grey band in (b) and (c) is the floor a ramped gain alone produces (section 6.4).

I THE THEILER EXCLUSION

I.1 The exclusion cap at the twenty-component configuration

The rule of appendix B.1 asks for at least the full span of a delay vector, so that two neighbours cannot be near each other merely because their windows overlap; the implementation caps that at 150 samples.

At the eight-component configuration the span is 76, so the cap never binds and every result in sections 5 to 7 is scored at 76. At the twenty-component configuration the span is 624 and the cap binds unconditionally, which is why table 9 marks that result exploratory.

Table 19 measures the cost of that departure.

Table 19: The cost of the exclusion cap. The twenty-component system re-scored at the capped exclusion and at the one the configuration asks for. Seven ranks, three seeds, the three observers with the lowest error, median over seeds; the error is against the measured PR^{pos} and is not comparable with table 6, which covers more observers.

r	measured PR^{pos}	MG at $W_T = 150$	MG at $W_T = 624$
2	2.00	1.74	1.81
5	5.00	4.16	5.15
8	8.00	6.28	6.71
11	11.00	9.54	10.26
14	14.00	9.81	9.37
17	17.00	14.40	14.19
20	19.99	15.92	16.60
mean absolute error		2.16	1.89
Spearman ρ		1.00	0.96

The ordering of the ranks, all that section 5.1 claims at this range, is nearly unaffected, a single inversion apart. The mean absolute error falls when the protocol is enforced, so the cap was not making the reported result look better than it is. The saturation above about six components is present at both settings and is therefore not an artefact of the exclusion.

The identifiability ratio is where the cap does bite. Its numerator is computed at $2E_{\max}$ with τ held fixed, so that span of 156 exceeds the cap while the denominator’s 76 does not. Every ratio in this paper is therefore 76 against 150, the reference band of section 6.3 and the grokking logs alike, so the comparison in section 7.1 is like for like; but it is not the fixed-exclusion quantity section 3.5 describes.

I.2 The Theiler exclusion and the dimension of a transient

Section 3.4 claims that recurrence is a requirement of this protocol and not of the estimator. The claim is testable on the excitation cases of section 5.1 by varying the Theiler exclusion W_T alone, from zero up to the value the frozen rule gives (table 20).

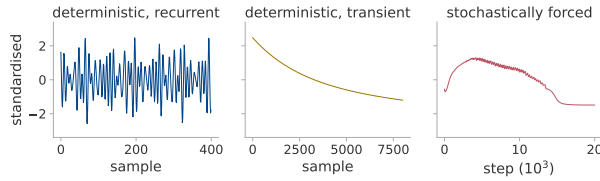


Figure 13: The three regimes as raw scalar logs. Each panel is the parameter norm, standardised to zero mean and unit variance as in Algorithm 1. Left, the constrained head of section 5.1 under fast quasiperiodic forcing, recurrent on a torus. Middle, the same system under plain gradient descent, transient over its whole record. Right, a transformer grokking modular addition, stochastically forced.

Table 20: The estimate against the Theiler exclusion, median over ranks, seeds, observers and windows. At $W_T = 0$ the transient returns the dimension of its own path, which is not the dimension of a regime (section 3.4); the exclusion removes the neighbours that value rests on. The fast recurrent case is unchanged under the same variation. Column 76 is the exclusion the frozen configuration asks for and the one every other result uses; the last removes the cap of 150 and applies the autocorrelation rule, some 1 700 samples here.

W_T	0	2	5	20	76	100	150	uncapped
recurrent, fast	4.83	4.82	4.82	4.82	4.80	4.80	4.80	4.80
recurrent, slow	2.06	3.02	3.22	3.27	3.27	3.27	3.28	3.27
transient	1.20	1.71	2.38	5.51	16.12	20.50	29.22	173.19

The value at $W_T = 0$ is the dimension of the path. On a uniformly sampled curve the nearest neighbours of a point are its immediate temporal neighbours, for which equation (3) returns 1.199; the transient returns that value in every cell. The excess over unity is the lattice bias of a deterministic sampling.

The exclusion removes the near neighbours, not the neighbours. The estimator fills its twenty slots at every exclusion; only the distances change. Before any exclusion the nearest neighbours of a point lie thousands of samples away under fast quasiperiodic forcing, but six samples away on the transient (figure 13). Excluding 150 samples therefore costs the fast forcing almost none of its close pairs and the transient all of them (figure 14). The distances the transient retains are then nearly equal, their log-ratio sum collapses, and the reciprocal in equation (3) diverges. An exponential decay with no optimiser and no data behind it reproduces both ends.

The asymmetry is general; the level is not. Under fast forcing $W_T = 0$ already returns r , because that case is not oversampled; it is the slow forcing that the exclusion repairs, its nearest neighbours lying some seven hundred samples away.

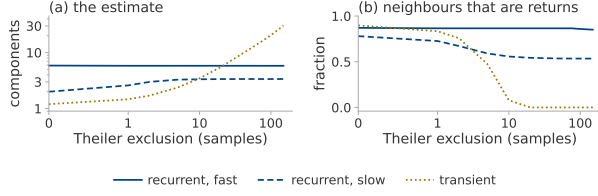


Figure 14: The exclusion costs a transient every close pair and a torus almost none. The excitations of table 20 on the parameter norm, median over ranks, seeds and windows. In (b) a neighbour counts as a return when it is no further away than the nearest neighbours an unexcluded search finds, so a fall to zero leaves the estimator nothing but distant points.

A recurrence-based diagnostic. The ratio $\hat{d}_{MG}(W_T=150)/\hat{d}_{MG}(W_T=20)$ is unity in both recurrent cases and above five on the transient, so it tests recurrence in the reconstructed space where the trend-crossing count tests only non-monotonicity of the series: the gap section 3.5 leaves open, which appendix K shows to matter on real runs. The principled version is older than either: a space-time separation plot (Provenzale et al., 1992) shows directly how neighbour distance depends on separation in time, and the exclusion should be chosen from one.

J THE TRAINING RUNS

Table 21 lists every training run the paper draws on. **T** is the one-layer transformer of Nanda et al. (2023): width 128, four heads, no layer normalisation, 226 816 parameters on modular addition and 228 608 on S_5 , trained in double precision with AdamW. **P** is the two-layer quadratic perceptron of Gromov (2023): width 500, 145 500 parameters at $p = 97$, trained by full-batch gradient descent with no regularisation.

Nanda et al. (2023) train at full batch and we use mini-batches. In the perceptron setting we take the tasks and the representability criterion of Doshi et al. (2024) but not their optimiser, running plain full-batch gradient descent on mean squared error instead. The S_5 task (Power et al., 2022) and its weight decay lie outside the configuration of Nanda et al. (2023). The two S_5 runs also step the optimiser twice per batch, so their effective learning rate and decay are about double the nominal ones.

Definition of memorisation and generalisation. The memorisation step is the first logged step at which the training accuracy reaches 95%; the generalisation step is the first at which the validation accuracy does. Neither requires the threshold to be held thereafter. The extended reruns of section 7.2 use the stricter rule

instead, requiring the threshold to hold over a sustained block.

Table 21: Training runs used in this paper. α is the training fraction, “final” the validation accuracy at the end of training. A dash in the generalisation column marks a run that had not generalised when training ended; such runs are censored observations, not established negatives.

run	task	arch	wd	α	steps	mem.	gen.	final	used in
<i>Trajectory sketch attached</i>									
mod_wd1	mod. add. $p=113$	T	1.0	0.30	20k	1470	13660	1.00	section 7.2
mod_wd1_s43	mod. add. $p=113$	T	1.0	0.30	20k	1580	6640	1.00	section 7.2
mod_wd1_s44	mod. add. $p=113$	T	1.0	0.30	20k	1410	3680	1.00	section 7.2
s5_wd1	S_5 composition	T	0.2	0.50	15k	1280	6685	1.00	section 7.2
mod_wd0	mod. add. $p=113$	T	0.0	0.30	20k	630	—	0.03	section 7.2
s5_wd0	S_5 composition	T	0.0	0.50	15k	1015	—	0.01	section 7.2
<i>Extended reruns</i>									
gropos_s0	mod. add. $p=113$	T	1.0	0.30	120k	1450	5070	1.00	section 7.1
lowdata15_s0	mod. add. $p=113$	T	1.0	0.15	120k	620	—	0.58	section 7.2
lowdata15_s1	mod. add. $p=113$	T	1.0	0.15	120k	630	106380	1.00	section 7.2
lowdata15_s2	mod. add. $p=113$	T	1.0	0.15	120k	660	—	0.01	section 7.1
lowdata20_s0	mod. add. $p=113$	T	1.0	0.20	120k	800	38380	1.00	section 7.2
wd0_s0	mod. add. $p=113$	T	0.0	0.30	120k	610	—	0.09	section 7.1
wd0_s1	mod. add. $p=113$	T	0.0	0.30	120k	630	—	0.07	section 7.1
p211_wd0	mod. add. $p=211$	T	0.0	0.16	200k	1800	—	0.04	section 7.2
<i>Perceptron, $p = 97$, full batch, no regularisation</i>									
a_add	$n + m$	P	0	0.5	100k	9230	11760	1.00	section 7.2
a_mul	$n \cdot m$	P	0	0.5	100k	9100	11560	1.00	section 7.1
a_sub	$n - m$	P	0	0.5	100k	9110	11540	1.00	recorded only
a_sq_sum	$n^2 + m^2$	P	0	0.5	100k	8440	10470	1.00	recorded only
a_sum_sq	$(n + m)^2$	P	0	0.5	46k	7350	8190	1.00	recorded only
x_mix_quad	mixed quadratic	P	0	0.5	100k	9600	—	0.01	figure 8
x_no_grok	$n^3 + nm^2 + m$	P	0	0.5	100k	9860	—	0.01	section 7.2
g_p1	$(4n_1 + n_2^2)^3$	P	0	0.5	100k	7470	12290	1.00	section 7.2
g_p1x	$+ n_1 n_2$	P	0	0.5	100k	10140	—	0.01	section 7.2
g_p2	$(2n_1 + 3n_2)^4$	P	0	0.5	100k	6470	7680	1.00	section 7.2
g_p2x	$- n_1^2$	P	0	0.5	100k	9730	—	0.02	section 7.2
g_p3	$(5n_1^2 + 2n_2^2)^2$	P	0	0.5	100k	5970	6600	1.00	section 7.2
g_p3x	$- n_2$	P	0	0.5	100k	9420	—	0.74	section 7.2

Four of the perceptron runs are repeated with the trajectory sketch attached, for the measurement of appendix H.1: **a_add** and **x_no_grok** from the table above, together with the label-matched pair **g_p1** and **g_p1x**. Each is run at full batch and again with mini-batches of 512. The full-batch runs reproduce the memorisation and generalisation steps of the rows above exactly and the mini-batch runs to within six logged steps, so appendix H.1 compares two noise levels on matched learning.

g_p3x ends far above the majority-class baseline of 1.3% while being provably outside the class of functions the architecture can represent.

K DESCENT AT THE EDGE OF STABILITY

Descent at the edge of stability (Cohen et al., 2021) is gradient descent at a fixed rate: the top Hessian eigenvalue sharpens until it reaches $2/\eta$ and then stays there, while the loss descends over long stretches and not over short ones. Such a run is deterministic and is not a plain transient, so an unforced trajectory could satisfy the conditions of section 3.4.

Setting. We run full-batch descent without weight decay on the quadratic perceptron of appendix J (ta-

ble 22). The log is written at *every* optimiser step, because the oscillation is the two-cycle of the unstable mode and its period is a few steps. The sharpness is measured by power iteration on Hessian-vector products, taking the largest *algebraic* eigenvalue, since the Hessian is indefinite early in training.

Table 22: Oscillation and recurrence in unforced descent at the edge of stability. Full-batch descent by learning rate, both seeds. The stability ratio is $\eta\lambda_{\max}/2$ over the second half of the run; *rises* is the fraction of post-transition steps on which the loss increases; *return* is the nearest neighbour surviving a 150-sample exclusion divided by the distance the trajectory travels during that exclusion, for which a two-frequency torus scores 0.008.

η	stability ratio	rises	return	outcome
1×10^5	0.239, 0.238	0.000	1.007	monotone
3×10^5	0.758, 0.745	0.000	1.007, 1.007	monotone
1×10^6	0.969, 0.971	0.495, 0.490	0.104, 0.079	at the edge
1.5×10^6	0.974, 0.970	0.500, 0.500	1.012, 1.011	at the edge
2×10^6	0.977, 0.978	0.500, 0.500	0.120, 1.013	at the edge
2.5×10^6	—, 0.988	—, 0.500	—, 0.034	one seed diverges
2.8×10^6	0.992, —	0.503, —	0.022, —	one seed diverges
3×10^6	—	—	—	diverges

Oscillation is not recurrence. Eight runs reach the edge and all eight oscillate, the loss rising on one step in two, but only five of them return close to states already passed through. The return column of table 22 separates those two groups by a factor of eight with nothing in between. The split is not a threshold: at $\eta = 2 \times 10^6$ one seed returns and the other does not, at 0.120 against 1.013, while their sharpness and their fraction of rising steps agree to three decimals.

Performance of the diagnostics. The band of section 6.3 rejects all eight runs at the edge, on the training loss and on the parameter norm alike, whether or not they recur, ρ_{ident} lying far outside it in both groups while the trend-crossing count, which tests non-monotonicity and not recurrence, passes every one of them. The estimate does not separate the runs that return from those that do not either: their ranges on the training loss overlap.

Dependence on the logging stride. Written at the stride every other full-batch log here uses, most of the phenomenon disappears (figure 15): over these eight runs the median trend-crossing count falls from 71 to 6. Whether a log looks like a transient is thus in part a property of how often it was written.

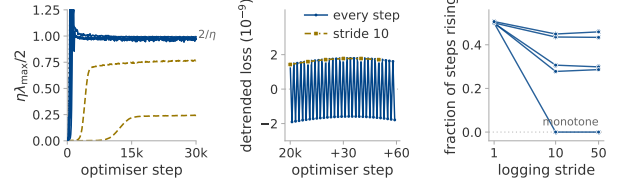


Figure 15: Full-batch descent reaches the edge of stability. (a) The stability ratio along training, one seed per rate. (b) Sixty consecutive steps at one rate at the edge, against the same series at a stride of ten: the two-cycle is removed by that stride rather than blurred by it. (c) The fraction of steps on which the loss rises, against the logging stride.

L THE CEILING: EMBEDDING DIMENSION AGAINST RECORD LENGTH

Two explanations were available for the ceiling of section 5.1: the embedding condition $E > 2d$ and the finite-record bound $2\log_{10} N$ (Eckmann and Ruelle, 1992). Each is insensitive to the variable the other names, so we sweep $E_{\max}$ at a fixed record length and the record length at fixed $E_{\max}$, over $r \in \{2 \dots 20\}$ with everything else frozen (table 23). Each record length is a *prefix* of one longer record rather than a resampling of a fixed span, leaving the delay lag its meaning in periods. Every trajectory is bit-identical to the published one over their common prefix.

Table 23: The ceiling follows neither the embedding condition nor the finite-record bound. Two sweeps, three seeds per cell. *Tracking* is the largest r at which the median estimate stays within one component of the truth; *level* is the median estimate at $r = 20$; *slope* is $d\hat{d}_{MG}/dr$ over $r \geq 8$, which is 1 if the estimator still tracks the rank and 0 if it has stopped.

$E_{\max}$ at $N = 8000$	10	14	20	28	40	56	
tracking	6.5	7.3	8.1	10.3	10.9	11.2	
$E_{\max}/2$ predicts	5	7	10	14	20	28	
level at $r = 20$	6.9	8.1	9.8	11.3	13.2	14.9	
slope	0.08	0.12	0.20	0.29	0.41	0.54	
N at $E_{\max} = 20$	10^3	$2 \cdot 10^3$	$4 \cdot 10^3$	$8 \cdot 10^3$	$1.6 \cdot 10^4$	$3.2 \cdot 10^4$	$6.4 \cdot 10^4$
tracking	6.7	8.3	8.5	8.1	8.9	9.5	9.5
$2 \log_{10} N$ predicts	6.0	6.6	7.2	7.8	8.4	9.0	9.6
level at $r = 20$	9.0	9.3	9.4	9.8	9.9	10.2	10.5
slope	0.22	0.17	0.17	0.20	0.19	0.19	0.22

Outcome. The level at high rank is set by the embedding: a sixty-fourfold increase in the record adds 1.5 components, whereas a sixfold increase in $E_{\max}$ adds eight. The tracking ceiling follows neither variable. It never passes 11.4, yet at a quarter of the record the rest of the study uses it still tracks to 8.3, where $2\log_{10} N$

allows only 6.6. A bound set by the record length cannot produce that, so Eckmann and Ruelle (1992) is not the explanation here. The embedding condition $E > 2d$ fails too in the form we used it: the ceiling rises by a tenth of a component per unit of $E_{\max}$ rather than by a half, so $E_{\max}/2$ happens to be right near $E_{\max} = 14$ and is wrong at both ends.

Fitting the level at high rank separates the candidates (table 24): neither hypothesis fits, with or without a free scale, and a form logarithmic in both variables absorbs the residual variation. That form has three parameters on sixteen cells and no held-out cell, so it describes the surface and does not explain it.

Table 24: Neither candidate explanation of the ceiling accounts for the measured level. Each form scored at $r \geq 16$ over the sixteen measured ($E_{\max}, N$) cells, in sample: no cell is held out, so the error falls with the parameter count by construction. The spread of the fitted quantity itself is 2.00 components.

form	parameters	root-mean-square error
$E_{\max}/2$, the embedding condition	0	8.73
$2\log_{10} N$, the finite-record bound	0	3.68
the pointwise minimum of the two	0	3.70
$0.56 E_{\max}/2$	1	3.41
$1.36 \cdot 2\log_{10} N$	1	2.45
the pointwise minimum, each scaled	2	1.08
$8.61 \log_{10} E_{\max} + 0.90 \log_{10} N - 5.31$	3	0.19

A purely linear statistic reproduces the whole dependence. Across the $E_{\max}$ sweep PR_{delay} rises over the same range as MG; across the record sweep it is flat; and a single logarithm in $E_{\max}$ fits it more tightly than any form fits anything else in this appendix. Most of the ceiling is therefore the number of components the delay window resolves *linearly*.

The unexplained tracking limit. The ceiling lies near eleven components for no reason we can give. It does not follow the measured effective rank, which equals r throughout, nor does any form we fitted account for it.

The identifiability ratio of section 3.5 does register it. At $r = 20$ the ratio stays above the band at every record length, so a longer record never makes an unidentifiable estimate identifiable; it falls into the band as $E_{\max}$ grows, at about the rank where tracking stops.

M TWO CONTROLS SEPARATING THE ESTIMATOR FROM ROUGHNESS

The number of components changes and the roughness does not. Both systems are a sum of four sinusoids. In

the first the four frequencies are rationally independent, so the path lies on a four-dimensional torus and $d_{\text{act}} = 4$. In the second they are harmonics of one irrational base frequency: a single phase fixes all four sinusoids at once, the path is a circle, and $d_{\text{act}} = 1$. The base frequency of the second is set by bisection so that the two have the *same* roughness. The schedule between them ramps the envelopes; every window a ramp reaches into is left unscored.

The roughness changes and the number of components does not. Five monotone observers $g(x) = x + ax^p$ are applied to one four-phase system. Each is invertible, so it may bend the recorded set but cannot add or remove a coordinate. Each also changes the local scale $g'(x)$ and with it the spacing of consecutive samples.

Table 25: Both controls, median over eight seeds. Above the rule, two systems matched in roughness and differing in the number of independent phases. Below it, one four-phase system under five invertible maps. PR_{delay} is the linear participation ratio of the delay covariance, included because it follows the roughness rather than the active dimension in both halves.

	d_{act}	estimate	roughness	PR_{delay}
four independent phases	4	3.92	0.091	2.20
one phase, four harmonics	1	1.11	0.091	4.10
identity	4	3.92	0.091	2.20
$x + 0.25x^3$	4	3.96	0.098	2.32
$x + x^3$	4	3.96	0.110	2.58
$x + x^5$	4	4.01	0.146	3.61
$x + 0.1x^7$	4	4.02	0.166	4.28

The tolerance was fixed before the runs, at a tenth of a component about four. Table 25 shows it holds.

Both controls run on a constructed system whose phases are known, so they say nothing about a training log.

Two findings count against the estimator. The linear participation ratio is higher on the circle than on the four-torus (table 25), so anyone treating it as a count of phases would order the two backwards. On the circle the estimate is also not geometric: every neighbour surviving the exclusion there is a sample from another pass around the circle. The local speed of the curve multiplies all of a point’s neighbour distances alike; speed cancels from the log-ratios of equation (3). The residue depends on the arithmetic of the base frequency: two of the eight seeds drew a frequency whose multiples cluster into near-duplicates; their estimate is reproducible from the fractional parts of those multiples alone, with no signal and no neighbour search.

N COMPUTE COST AND CODE AVAILABILITY

Compute cost. Everything here runs in a few hours on one consumer GPU and one desktop CPU. The largest GPU item is the edge-of-stability experiment of appendix K, which takes half an hour on a single T4; the analysis is the more expensive half and runs entirely on CPU, a few core-hours per experiment.

Cost of the trajectory sketch. At each logged step the sketch keeps 4096 floats against the full parameter vectors of appendix J: a run whose raw trajectory would occupy 88 megabytes occupies 2.5. We train the same configuration with and without the sketch attached, alternating the order so that warm-up cannot favour either. At a logging stride of ten steps the sketch adds about six per cent to the training time. `sketch_cost.py` records the measurement.

Code availability. All code, configurations and result files behind every number and figure in this paper will be released publicly on acceptance. The figures regenerate from the committed result files with a single command.